NTU BRIDGE DESIGN COMPETITION 2023



Technical proposal

By

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Section 1: Proposed Bridge Design and Alignment

1.1 Proposed Bridge Design

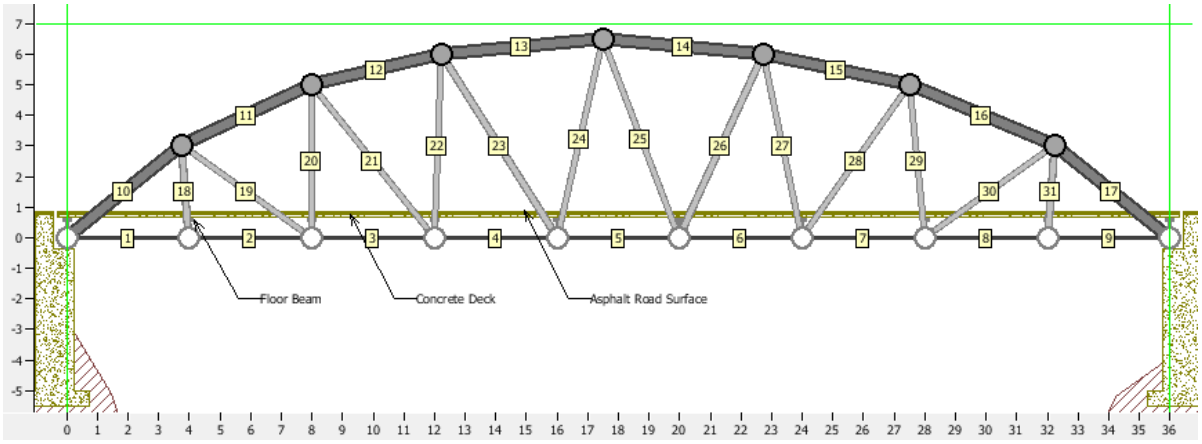


Figure 1. Design of bridge

The proposed bridge design is as shown in Figure 1., The bridge will have a total span of 36 meters and height of 7 meters. The structure of the bridge can be split into three parts: arch frame, deck, and truss. Members 1-9 form the deck, these are 70x70 High-strength low-Alloy solid steel bars; Members 10-17 form the main arch frame, these are 200x200 High-strength low-Alloy steel hollow tubes; Members 18-31 form the truss, these are 100x100 carbon steel hollow tubes, with members 18,23,26,31 being 120x120 carbon steel hollow tubes. The members of the main arch frame resists compression force, while the members forming the deck resist tension force. Truss members resist both tension and compression.

1.2 Alignment of Bridge and Road

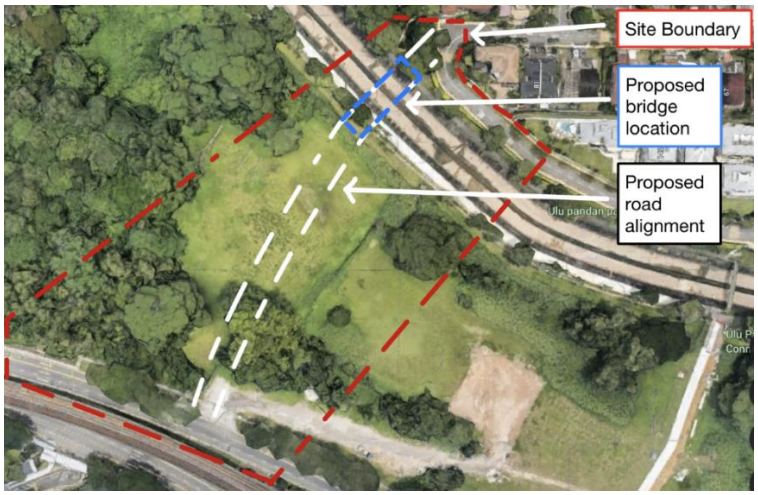


Figure 2. The proposed alignment of bridge and road

The proposed alignment of bridge and road is shown in Figure 2. The junction of the south-west connection point will be 90 degrees, as suggested by Land Transport Authority (LTA) [1]. The road leading to the bridge will be slightly curved to fit the terrain. Lastly, the road leading from the bridge to the north-east connection point will be straight. As the junction here is not 90 degrees, special traffic measures need to be taken into consideration.

1.3 Slope Design of Bridge

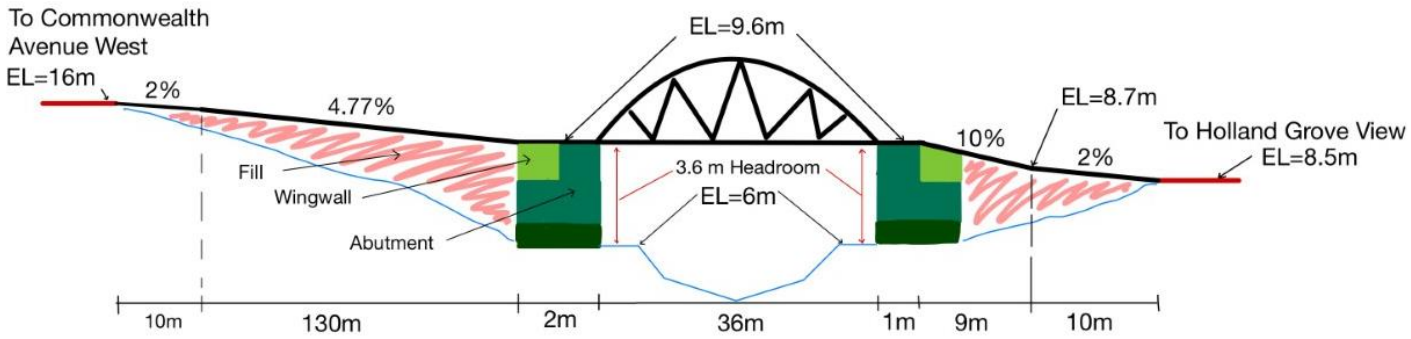


Figure 3. Slope design of bridge (not drawn to scale)

The slope design of the bridge and road must adhere to the standards of the Code of Practice provided by LTA. The gradient of slope of a road must not exceed 10%. Moreover, if a road steeper than 1:30 approaches a junction, an intermediate 10-meter road of 1:50 must be provided. [1]

The south-west connection point of the proposed bridge will have an elevation of 16 meters. As a headroom of 3.6 meters is needed on both sides of the bridge, the bridge will have an elevation of 9.6 meters. The road leading to the bridge will have a slope gradient of 2% followed by 4.77% before meeting the horizontal of the bridge.

The north-east connection point will have an elevation of 8.5 meters. From the bridge, a horizontal road of 1 meter, followed by a 9-meter road with a 10% slope and 10 meter road of 2% slope is proposed. According to the standard code of practice, the 9-meter slope must be constructed with rigid pavement, since it is 1:10. [1]

Section 2: Method Statement and Construction Sequence

2.1 Summary of the Construction Sequence [2]

Construction sequence	
Phase 0: Design Phase and Approval Phase	
Phase 1: Preliminary works	
1.1 Setting Out	
1	A site investigation and survey to assess the geotechnical conditions, check for any underground utilities and conduct environmental impact assessments.
2	Utility diversion and protection to ensure that existing underground utilities such as gas, water, and electrical lines are not affected by the construction work.
3	The construction site must be cleared of existing structures, debris, and vegetation.
4	Access roads to the site need to be constructed, and the ground levelled to ensure a suitable base for construction work.
5	Installing erosion and sediment control measures to prevent soil runoff into nearby water bodies.
6	Mapping out gridlines and erecting site hoarding, marking contract boundaries and installation of site hoarding and demarcating service lines.
7	Installation of acoustic hoarding barriers, noise monitoring devices, netting and shielding.

8	Establishing a site office, storage and workshop areas, and site welfare facilities for workers.
1.2 Offsite Preliminary Works	
1	Procurement of construction materials.
2	The fabrication of bridge components such as steel girders and concrete deck slabs.
Phase 2: Foundation and Abutments	
2.1 Pile Foundation	
1	Install piles using a boring machine or a hydraulic hammer at the location of the pile.
2	Place a casing and reinforcing cage in the void and cast steel-reinforced concrete to form the pile.
3	Construct a reinforcement pile cap to transfer the load of the bridge to the pile.
2.2 Construction of Abutments	
1	Excavate the soil to the required depth to create the foundation for the abutment.
2	Install a grillage of steel reinforcements for the abutment.
3	Erection of vertical formwork for abutment with shores and bracing.
4	Fill formwork with concrete and compact by vibrating.
5	Remove the shutters, cure the concrete, treat exposed surfaces, and apply a waterproof membrane (bitumen coating applied by brush or spray) on the faces retaining soil fill.
6	Laying of the drain (small pipe with granular surround) behind abutments.
7	Place and compact granular fill behind abutments and wing walls to the road formation level.
2.3 Bearing Installation [3]	
1	Prepare the top of the abutment for bearing placement. (As shown in Figure 10 in Appendix)
2	Install the bearing devices supporting the bridge's superstructure and allowing for movement and rotation.
3	Place neoprene or elastomeric pads on top of the bearings to accommodate the movement of the bridge.
4	Place the steel beams or girders on top of the neoprene or elastomeric pads and secure them to the bearings.
Phase 3: Construction of the bridge	
3.1 Arch Frame	
1	Assembly of arch segments using bolts or welding then install temporary support and tie bars.
2	Lifting (Figure 13 in Appendix), alignment and welding of the arch segments to form a rigid structure then install arch bearings to support the arch frame.
3.2 Bridge Deck	
1	Precast concrete beams are placed atop temporary supports at abutments.
2	Install steel deck framing along floor beams.
3	Place precast concrete deck on top of steel deck framing.
4	Finish with asphalt road surface above of precast concrete deck.

2.2 Foundation and Soil Work

A crawler excavator or backhoe excavator is used, which is favourable on sites with uneven terrain. A deep foundation is chosen for this project. It is necessary where the bearing capacity of the surface soils is insufficient to support the loads imposed and so they are transferred to deeper layers with a higher bearing capacity [4]. The soil condition may be poor and not stable enough as it is located near a canal and drainage system. A temporary casing is needed to protect the soil from collapsing into the borehole during drilling. Site investigation and soil analysis must be done.

A bored pile retaining wall is used. A bored pile is more suitable due to the vibration or noise levels generated by a pile driver are not acceptable due to the concerns of nearby residents. Earth Retaining Stabilizing Structure (ERSS) design is used to maintain the shape of excavation during construction and earth filling. [5]

2.3 Construction of Arch Frame

The bridge's arch frame has a length of 36m. Vehicles exceeding 13m are not allowed on LTA roads, so the arch frame span must be divided and kept under 12m. It is suggested to deliver each member separately to the site. On the site, the frame is welded together, and the members are connected through bolts instead of screws.

2.4 Assembly of Truss member

At the point where the arch frame components connect, gusset plates are installed. Prior to the erection of the arch frame, the truss members are connected to the arch frame on the gusset plates. [6] The frame's dimensions (Figure 13 in Appendix) are small enough to be assembled on-site.

2.5 Erection of Arch Frame and Equipment Used

Based on the software, the weight of the arch frame without decks is 3.639 tons. A rented or second-hand crawler crane (model Kobelco 7065-2 [7]) is sufficient as the maximum weight of the arch frame, which is about 3.7 tons at 22m (The crane stays away from the canal at least 3 meters for safety purpose). The crane can carry a maximum load of 6.3 tons at this distance (Figure 15 in Appendix). Thus, the factor of safety is 1.7 (>1, safe range). The crane is installed on a tracked vehicle with metal tracks that increase the crawler crane's stability on the ground, suitable for all types of grounds and uneven terrain. Temporary propping systems are installed at the base of the frame to ensure the frame's stability.

2.6 Prefabricated Steel Deck Framing and Lateral Stability

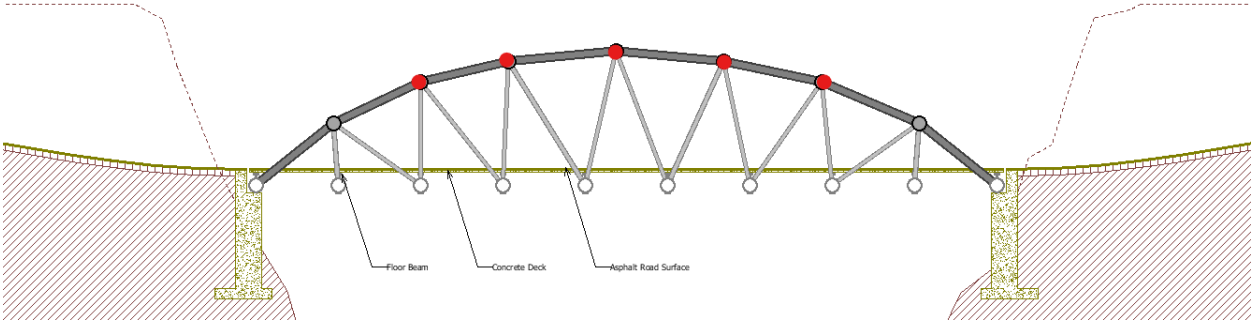


Figure 4: Arch frame without deck and its location of lateral truss and bracing connection points (Red).

The 4m x 7m deck framings are constructed off-site. It's worth mentioning that specific road approvals are necessary when the items being transported have a width of more than 3m. The steel deck must be moved during off-peak hours (0700-0930 and 1700 to 2130), even if the LTA has granted authority. As a result, this must be accounted for in the construction schedule and planning.

The steel deck frames can be lifted and constructed on-site using a crawler crane (Figure 14 in Appendix). A gusset plate [6] is prefabricated on the steel deck frames to attach the arch frame to the framing. Additionally, lateral stability must be considered, with the positions for lateral truss and bracing connections shown in red in Figure 4. Bracing must be installed throughout the prefabricated steel deck to account for the bridge's lateral stability and reduce global buckling for the main arch and other horizontal loads such as wind or unintended vehicle loading. Finally, the precast concrete deck can be placed on top of the steel deck framework, and asphalt concrete can be poured over it. This completes the superstructure of the bridge.

Section 3: Site Constraints and Proposed Solution

3.1 Traffic Management

To facilitate logistics and delivery, traffic management is required. The construction site can be accessed from both the south and north side of the construction site. From the south, the site is accessible from Commonwealth Avenue West, a 3-lane one-way road. From the north, the site is accessible from Holland Grove, a two-way road.

The south and north connection point will be both junctions, therefore, control measures are needed:

- Consult relevant authorities for traffic safety requirements for the locations.
- Provide traffic signboards and stop signs where necessary according to Code of Practice for Traffic Control at Work Zone [8].
- Provide traffic controller to facilitate the entry and exit of vehicles at site entry and the crossing of pedestrians.

For the south construction site, signboards for the new proposed entrance for pedestrian's walkway is also displayed. For exit from the junction, a stop sign is placed (Figure 5). Clearance of 3m width and 3.6m headroom for the park connector (Figure 12 in Appendix) to remain accessible during all construction stages.

On the north construction site, the Holland Grove will be split into two temporary roads (two-way) (Figure 6) as the construction site blocks the end of both roads, sign boards indicating the roadblock ahead and alternative pathways (green or brown road) will be placed at junctions. The entry and exit of vehicles at site will only be through the brown road in Figure 6, the traffic controller will ensure no pedestrians are crossing when vehicles enter and exit. The road lines are to be repainted after construction and the car parking spots on brown road next to houses are to be strictly prohibited during construction period (Figure 11 in appendix). Clearance of 2m width and 3.6m headroom for the park connector to remain accessible during all construction stages.

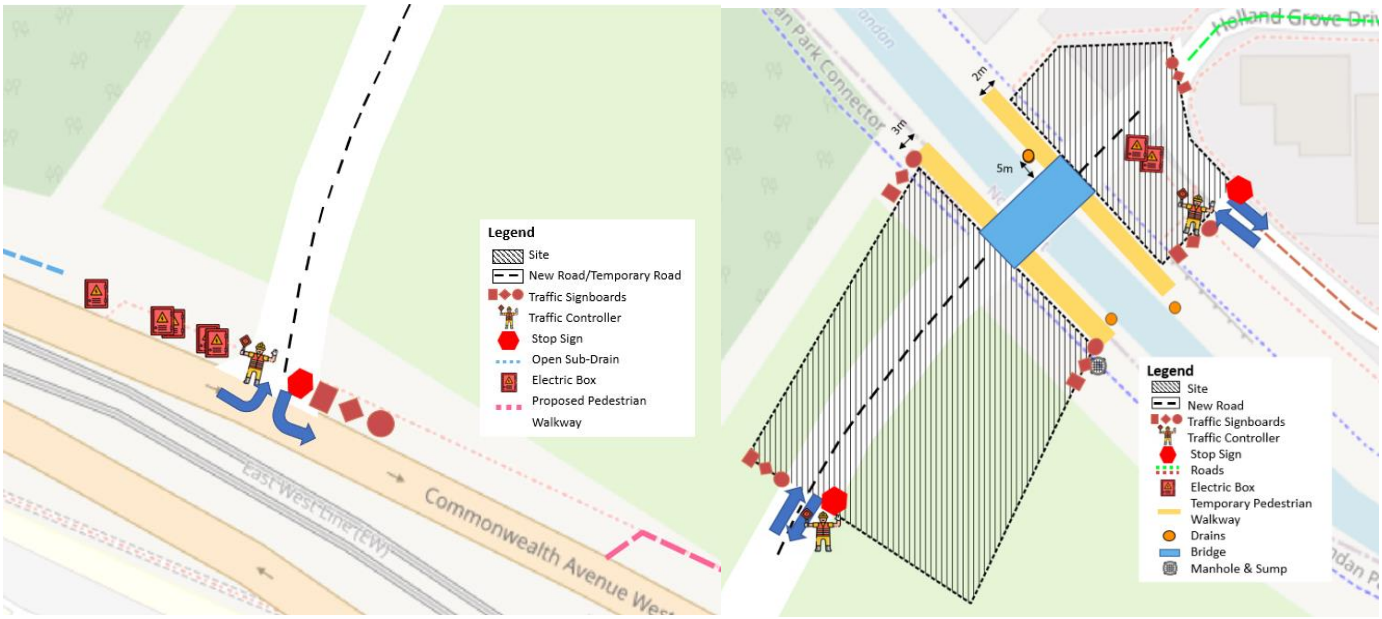


Figure 5. Traffic plan for South (Left) and North (Right) connection points.

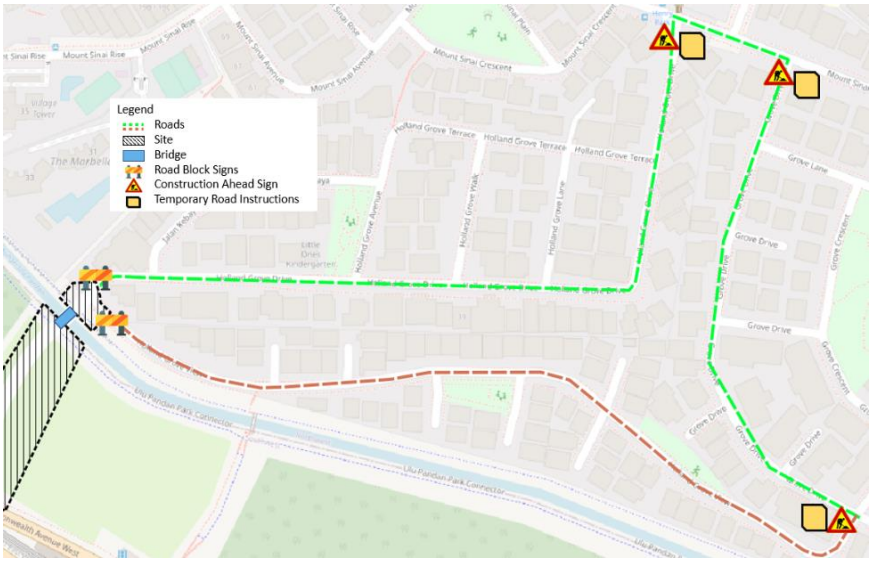


Figure 6. Proposed temporary roadblocks and road diversion

As crawler cranes are used, it is important to note that they are big and bulky vehicles and are not suitable to be transported as it is. During construction, parts of the crawler crane are to be transported separately to the site via trucks and to be assembled on site. Once the work is done, the crawler crane needs to be dismantled and transported out separately.

The unloading of materials and assembly of the crawler crane is to be done only within the site boundary. The site is surrounded by steel plate barriers for safety purposes and to avoid disruption caused to nearby facilities.

3.2 Site Procession - Hoarding Plan

Assuming a temporary road built from Commonwealth Avenue West to the South site, a 10m wide access is opened. Another 6m wide access is opened at Holland Grove at North site. Site access, site office, and storage area are shown in Figure 7.

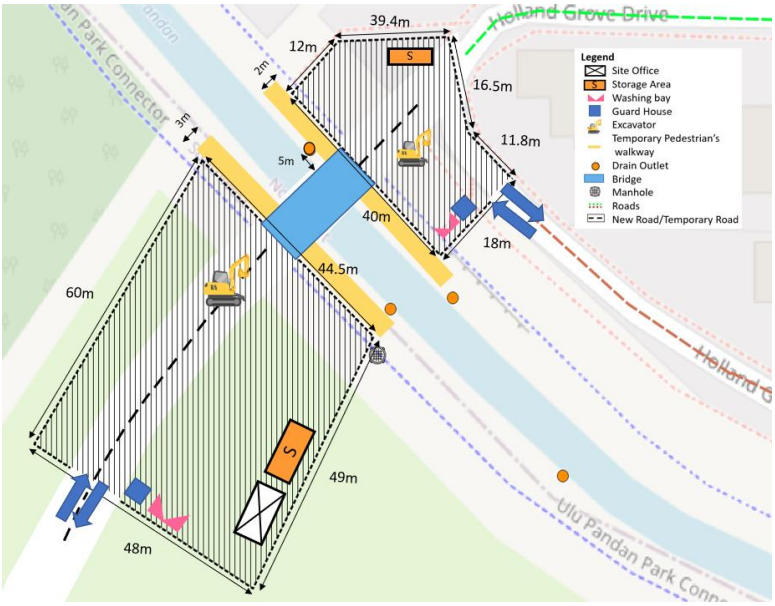


Figure 7. Proposed site plan

3.3 Public Sewerage System

There are 4 sub-drain outlets, a manhole and sump at the canal. Based on the Protection of Public Sewerage System by Public Utilities Board (PUB) [9], sewer setback net clearance for a sewer diameter 150-600mm, depth>5m is 2.0m from the outer edge of any structure to the centerline of sewer pipe. Therefore, the roads connecting to the bridge must be constructed at least 2m away from the manhole and drainage (Figure 5). We are required to engage a PUB registered CCTV contractor to carry out pre-construction and post-construction inspections.

3.4 Existing Lamp Post and Overground (OG) Box

There are 7 overground boxes affected by the construction, 5 at South and 2 at North (Figure 5). The ones at South may not need to be relocated, however, permission/guidelines from authority are needed before construction. 2 lamp posts at North would have to be relocated as well. For the new connecting road and bridge, street lights would need to be installed for the safety of night driving (Figure 9 in Appendix).

A Professional Engineer (PE) shall forward the street plan proposal with a cut-out plan of the affected streetlights, lighting control boxes and proposed positions with the inventory ID (e.g., lamp post number, OG box number) and road name attention to Lighting & Facilities Department. The PE shall also co-ordinate with all other technical agencies on whether the new lamp posts' locations are acceptable to all parties. The developer/owner shall bear the cost of the lamp posts relocation. The proposed relocation of the lamp posts shall comply with LTA's requirement. [1] In replacement of lamp post, temporary solar powered LED street lights will be used throughout the construction stage for the neighborhood to ensure safety of all residents and construction workers.

Section 4: Environmental impacts and measures

The construction shall be planned with sustainability in mind and be monitored throughout the construction process.

4.1 Trees in Green Space

As shown in Figure 8, the proposed road alignment spans across several trees in the green space. The trees will be transplanted to a nearby suitable greenspace. The detailed procedures of the transplanting need to be consulted with a certified arborist. The construction may only proceed after detailed analysis and approval by the Land Transport Authority (LTA) and National Parks (NPARKS). It is also recommended to plant trees in the green space nearby in an effort to offset the carbon of the project and conserve the biodiversity.



Figure 8. Relocation of Trees

4.2 Materials

- Efforts to reduce embodied carbon in materials are to be made to increase the sustainability of the project. For example, green concrete made from recycled materials such as washed copper slag, recycled carbon dioxide and ground granulated blast furnace slag (GGBS) can be used to construct the bridge decking [10].
- Waste from the project should be minimized by good quantity surveying before the start of construction. Construction waste should be handled according to the LTA Guidebook for Best Environmental Practices [11].
- Using Reclaimed Asphalt Pavement (RAP) which contains aggregate and bitumen that can be reused for asphalt production, keeping asphalt materials in a circular economy. [10] Plastic-bituminous asphalt roads could be used in the future after specifications for quality assurance and control being prepared. [12]
- Alternative solutions like inflatable dams and soil netting could be used as they are gentler on local ecosystems compared to sandbags. [13]

4.3 Equipment

- Usage of suitable fuels can increase efficiency, and therefore reduce running time and costs[14].
- Equipment must be switched off when not in use to reduce fuel consumption and carbon emissions.
- Usage of electric or hybrid construction machinery, therefore reducing on-site fuel consumption, minimizing energy loss, and decrease in air and noise pollution as compared to diesel-powered machines. Thus, can further reduce carbon emission. In situations where the use of electric equipment is not feasible, consider using diesel engines that meet EU Stage IIIA / US Tier III / JPN Tier III standards or better, to mitigate the emission of air pollutants (e.g. PM2.5) from construction equipment. [15]

4.4 Site Office

- Before the site office is set up, green considerations regarding the design of the office, e.g., use of alternative renewable energy sources like solar power and procurement of equipment should be examined. [15]

- During the procurement of office logistics, contractors should opt for fit-out amenities and furniture that could be redeployed to other sites after the project is completed. [15]
- Carbon accounting must be done and submitted to LTA for data analysis to track resources consumption monthly. [15]
- Usage of recycled water (for non-potable usage) for wash bay, boot wash and dust control. [15]

Section 5: Other Considerations

5.1 Site Safety

Site safety needs to be ensured with reference to the LTA construction safety handbook [16], with special attention to the following clauses: Housekeeping, Lifting Operations, Machinery Guarding, Working At Height, and Personal Protective Equipment.

Moreover, several actions are considered to ensure site safety.

1. Toolbox Meeting

   It will be held every morning before commencing work to brief the workers on daily activity plans so that everyone on site is aware of the whole site's activities, hazards, and safe work practices.

2. Daily inspections

   PPE inspections, checking, and verification of the on-field compliance including Safety Standards, working conditions, and other critical areas of the worksite. Housekeeping programs must be done, the checklist of tidiness and cleanliness on site is prepared.

5.2 Soil Improvement

Strengthening the surrounding soil through grouting to the ground to avoid excess settlement for the abutment. Grouting is relatively costly and so wastage must be controlled. The stability of slopes can be increased to good effect by anchoring them into the bedrock below the probable shear surface.[13] This is to ensure that heavy machinery can be placed on top of the slope, e.g., excavator and crane.

Another method will be chemical stabilizations which can reduce permeability of the soils, improve shear strength, increase bearing capacity, decrease settlement, and expedite construction. Commonly used chemicals for soil stabilization like lime, cement, and fly ash could be used. [17]

At North site, there are many low-rise buildings (1-2 stories) serving as residential area. These buildings are sat on shallow foundations, which are sensitive to settlement. Strengthening of the surrounding soil or retaining wall is needed, and settlement monitoring devices are required to track the settlement of the soil before, throughout, and after the construction process.

5.3 Noise and Dust Control

At North site, there is a housing area. Noise cancellation methods like acoustic hoarding barriers are installed before construction works to minimize the disruption to the surrounding residents and biodiversity in the green space, reducing noise pollution. As mentioned in Section 2.2, bored piles are used rather than driven piles to minimize vibrations and soil movements, minimizing noise pollution and the influence on adjacent buildings. Noisier activities like piling, concreting, and demolition are restricted to daytime (after 3 pm). Besides, noise monitoring devices and vibration monitoring devices are also used to ensure both limits are maintained throughout the whole construction, based on Table 1.

For concerns of dust pollution, effective measures such as water sprinklers/spray, shielding, netting, covers or hoarding for aggregate and sand storage would be taken to minimize dust pollution caused by construction works. The netting or barriers on the scaffolding of the construction site shall be of suitable height for effective containment of dust and debris. All construction debris should be properly stored and removed for disposal quickly and should not be left to accumulate at the site.[14]

Construction Noise Control

Maximum permissible noise levels for construction work commenced on or after 1 October 2007

	Mondays to Saturdays			Sundays and public holidays		
Types of affected buildings	**7am - 7pm**	**7pm - 10pm**	**10pm - 7am**	**7am - 7pm**	**7pm - 10pm**	**10pm - 7am**
(a) Hospital, schools, institutions of higher learning, homes for aged sick, etc	60 dBA (Leq* 12 hrs)	50 dBA (Leq 12 hrs)		60 dBA (Leq* 12 hrs)	50 dBA (Leq* 12 hrs)	
	75 dBA (Leq 5 mins)	55 dBA (Leq 5 mins)		75 dBA (Leq 5 mins))	55 dBA (Leq 5 mins)	
(b) Residential buildings located less than 150m from the construction site	75 dBA (Leq 12 hrs)	65 dBA (Leq 1 hr)	55 dBA (Leq 1 hr)	75 dBA (Leq 12 hrs)	-	
	90 dBA (Leq 5 mins)	70 dBA (Leq 5 mins)	55 dBA (Leq 5 mins)	75 dBA (Leq 5 mins)	55 dBA (Leq 5 mins)	
(c) Buildings other than those in (a) and (b) above	75 dBA (Leq 12 hrs)	65 dBA (Leq 12 hrs)		75 dBA (Leq 12 hrs)	65 dBA (Leq 12 hrs)	
	90 dBA (Leq 5 mins)	70 dBA (Leq 5 mins)		90 dBA (Leq 5 mins)	70 dBA (Leq 5 mins)	

Table 1. Maximum permissible noise levels for construction work. [18]

5.4 Notice to nearby residents

As the project will be very close to the housing area, a notice will be sent to the residents in the area before the commencement of the construction project. This letter should include information such as the benefits of the project, the estimated start and end date of the project, roadblocks, pedestrian walkways, impacts on residents or road users, etc.

Appendix

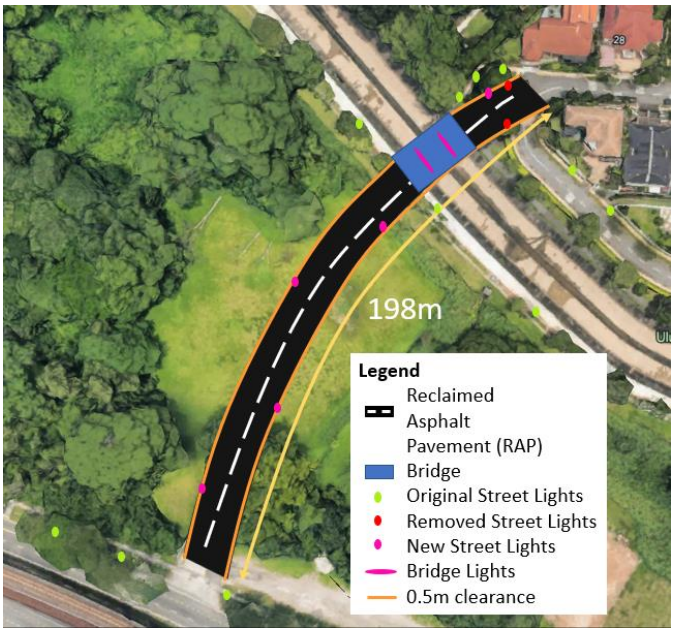


Figure 9. Proposed Road Plan



Figure 10. Elastomeric Bearing



Figure 11. Cars parking next to houses to be prohibited during construction stage



Figure 12. Pedestrian Temporary Walkway

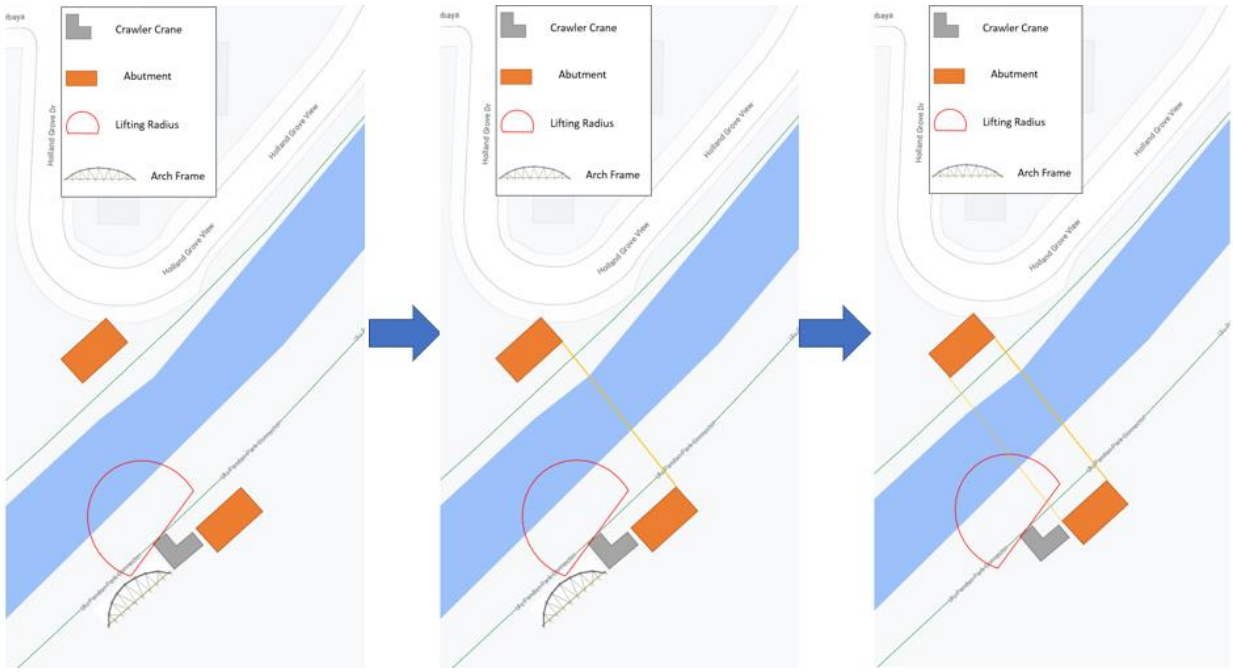


Figure 13. Assembly of Arch Frame

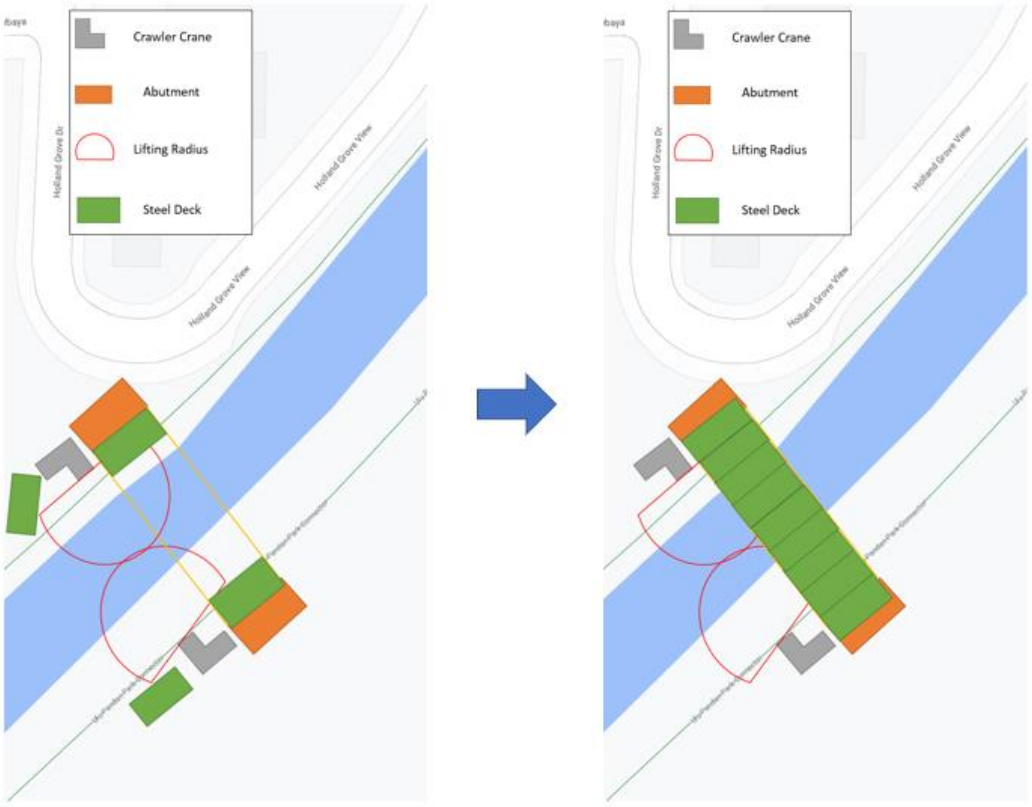


Figure 14. Installation of steel deck

Crane Boom Lifting Capacities

21.4(9.8+5.8+5.8)tonカウンタウェイト装着/クローラ拡張・全周 単位:ton

ブーム長さ m(ft) / 作業半径(m)	9.1 (30)	12.2 (40)	15.2 (50)	18.3 (60)	21.3 (70)	24.4 (80)	27.4 (90)	30.5 (100)	33.5 (110)	36.6 (120)	39.6 (130)	42.7 (140)	45.7 (150)	48.8 (160)	51.8 (170)	54.9 (180)
3.1	65.0	65.0/3.6m														
4.0	65.0	65.0														
4.1	65.0	65.0	63.1/4.8m													
4.5	59.1	59.1	59.0	56.5/4.7m												
5.0	52.6	52.5	52.3	52.1	47.9/5.0m											
5.5	45.1	45.1	45.1	45.1	44.7	41.3/5.7m										
6.0	39.4	39.3	39.2	39.1	39.0	39.0	35.9/6.3m	31.9/6.8m								
7.0	31.1	31.0	30.9	30.8	30.8	30.7	30.6	30.6	28.4/7.3m	25.4/7.9m						
8.0	24.8	25.6	25.4	25.4	25.3	25.2	25.1	25.1	25.0	24.9	23.0/8.4m	19.8/8.9m				
9.0	18.9	21.7	21.5	21.5	21.4	21.3	21.2	21.2	21.0	21.0	20.9	19.8	19.3/9.4m			
10.0	17.7/9.2m	18.8	18.7	18.6	18.5	18.4	18.3	18.2	18.1	18.1	17.9	17.9	17.8	17.7	16.3/10.5m	13.2/11.0m
12.0		13.7/11.9m	14.6	14.5	14.4	14.3	14.2	14.2	14.0	14.0	13.9	13.8	13.7	13.6	13.5	13.2
14.0			12.0	11.9	11.7	11.6	11.5	11.5	11.3	11.3	11.1	11.1	11.0	10.9	10.7	10.6
16.0			11.0/14.5m	10.0	9.8	9.7	9.6	9.5	9.4	9.3	9.2	9.2	9.0	8.9	8.8	8.7
18.0				9.0/17.1m	8.4	8.3	8.2	8.1	8.0	7.9	7.8	7.7	7.6	7.5	7.3	7.2
20.0					7.4/19.5m	7.2	7.1	7.0	6.8	6.8	6.6	6.6	6.5	6.4	6.2	6.1
22.0						6.3	6.2	6.1	6.0	5.9	5.8	5.7	5.6	5.5	5.3	5.2
24.0						6.2/22.4m	5.5	5.4	5.2	5.2	5.0	5.0	4.8	4.7	4.6	4.5
26.0							5.1/25.1m	4.8	4.6	4.6	4.4	4.4	4.2	4.1	4.0	3.9
28.0								4.3/27.7m	4.1	4.1	3.9	3.8	3.7	3.6	3.5	3.3
30.0									3.7	3.6	3.5	3.4	3.3	3.2	3.0	2.9
32.0									3.6/30.3m	3.3	3.1	3.0	2.9	2.8	2.6	2.4
34.0										3.0/33.0m	2.8	2.7	2.6	2.4	2.2	2.0
36.0											2.4/35.6m	2.4	2.3	2.1	1.8	1.7
38.0												2.0	1.9	1.7	1.5	1.4
40.0												1.9/38.3m	1.6	1.4	1.2	1.1
42.0													1.4/40.9m	1.2		
ロープ掛数	10	10	10	9	8	7	6	5	5	4	4	3	3	3	3	2

＊上表中の太線で囲まれた部分は、ブーム等の強度によって定められた値です。

Working Range

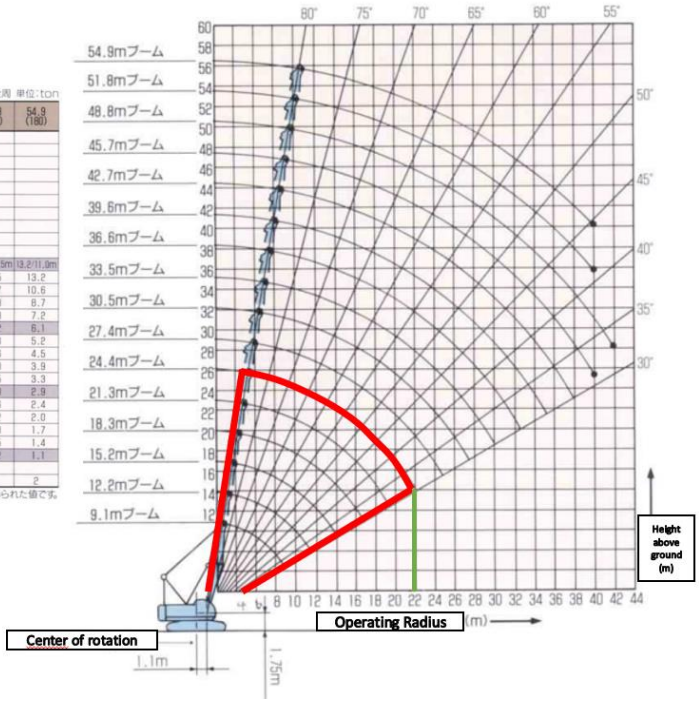


Figure 15. Boom Lifting Capacities and Working Range of KOBELCO 7065-2 Crawler Crane

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